

PRODUCT DOCUMENTATION

# Work About Work

## Calculator Methodology Sheet

|               |                                                                                 |
|---------------|---------------------------------------------------------------------------------|
| Document type | Calculator Methodology and Assumptions                                          |
| Version       | 1.1 (admin load changed to hours input; bar colours updated; em dashes removed) |
| Status        | Draft                                                                           |
| Companion to  | Framework Document v1.1                                                         |
| Audience      | Technical (developers, designers) and Business (CFO, ops leads)                 |

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### Purpose of This Document

This document explains the arithmetic, assumptions, and logic behind every number produced by the Work About Work calculator. It serves two audiences simultaneously:

- **Business readers** (CFOs, ops leads, senior stakeholders) who need to trust the outputs before acting on them
- **Technical readers** (developers, designers) who need to implement or modify the model accurately

*Every assumption is stated explicitly. Where a choice was made conservatively, that is noted. All inputs are configurable: the calculator is a business case generator, not a forecast.*

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### The Four Inputs

The calculator takes four user-adjustable inputs. Each is described below with its range, default value, and rationale.

| Input     | Variable  | Range | Default | Rationale                                                                              |
|-----------|-----------|-------|---------|----------------------------------------------------------------------------------------|
| Team size | team_size | 3–500 | 25      | Covers small squads to large departments. Above 500, org-level interventions dominate. |

|                                          |              |            |        |                                                                                                                                                                                                                               |
|------------------------------------------|--------------|------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Avg. fully-loaded salary                 | avg_salary   | €30K–€300K | €80K   | Fully-loaded: base + benefits + employer costs. €80K reflects a mid-market European knowledge worker.                                                                                                                         |
| Admin and coordination load (hours/week) | admin_hrs    | 2–28 hrs   | 14 hrs | Converted internally to percentage: $\text{admin\_pct} = (\text{admin\_hrs} / 40) * 100$ . Default of 14 hrs = 35% of a 40-hour week. Based on the 16.4 hr/week Microsoft/APQC benchmark; set conservatively below benchmark. |
| AI adoption rate                         | adoption_pct | 0–100%     | 50%    | Proportion of the team with consistent AI workflows in place. 50% reflects early majority adoption.                                                                                                                           |

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### Fixed Constants and Why They Are Fixed

Two constants are baked into the model and not exposed as sliders. Both are deliberate choices: changing them would require a methodology revision.

| Constant               | Value | Variable name  | Rationale                                                                                                        |
|------------------------|-------|----------------|------------------------------------------------------------------------------------------------------------------|
| Working weeks per year | 48    | weeks_per_year | Excludes approximately 4 weeks of annual leave and public holidays. Conservative baseline for knowledge workers. |
| Working hours per week | 40    | hours_per_week | Standard contracted hours. Overtime is excluded to avoid inflating the baseline.                                 |

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### The Reclaim Coefficient

The most important assumption in the model, and the one most likely to be challenged, is the reclaim coefficient: the proportion of admin hours that AI adoption can realistically eliminate.

**Value used: 0.60 (60%)**

This figure is derived from the three efficiency gain ranges in the Framework Document:

| Category                                         | Research range | Midpoint used | Source                                       |
|--------------------------------------------------|----------------|---------------|----------------------------------------------|
| Meetings and follow-up                           | 25–30%         | 27.5%         | Microsoft 365 telemetry; APQC                |
| Internal communications                          | 30–40%         | 35%           | Noy and Zhang (MIT)                          |
| Information retrieval                            | 25–35%         | 30%           | HBS: Jagged Technological Frontier           |
| <b>Weighted average (rounded conservatively)</b> |                | <b>~60%</b>   | Applied to admin hours only, not total hours |

***Important:** The reclaim coefficient applies only to the admin/coordination portion of time (admin\_pct), not to total working hours. A team spending 35% of their time on admin, with 60% of that reclaimable, recovers 21% of total working capacity, not 60%.*

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## The Calculation Logic: Step by Step

All output metrics are derived from the four inputs and two constants. Note: admin\_hrs (hours input) is converted to admin\_pct before use in the formula chain.

### Step 0 — Convert admin hours to percentage (Change 001)

$$\text{admin\_pct} = (\text{admin\_hrs} / \text{hours\_per\_week}) * 100$$

Converts the hours-based user input to a percentage for use in all subsequent calculations. At default of 14 hrs:  $(14/40)*100 = 35\%$ .

### Step 1 — Cost per hour

$$\text{cost\_per\_hour} = \text{avg\_salary} / (\text{hours\_per\_week} * \text{weeks\_per\_year})$$

Derives an hourly rate from the annual salary. Used to convert all time metrics into cost.

### Step 2 — Total team hours per year

$$\text{total\_hours} = \text{team\_size} * \text{hours\_per\_week} * \text{weeks\_per\_year}$$

The theoretical productive capacity of the team before any deduction.

### Step 3 — Admin hours lost per year

$$\text{admin\_hours} = \text{total\_hours} * (\text{admin\_pct} / 100)$$

The share of total capacity consumed by coordination overhead. This is the core defect quantity.

### Step 4 — Annual cost of coordination overhead [OUTPUT 1]

$$\text{admin\_cost} = \text{admin\_hours} * \text{cost\_per\_hour}$$

Translates the time defect into a monetary figure. This is the headline number in the tool.

Step 5 — Hours per person per week [OUTPUT 2]

$$\text{hours\_pp\_week} = (\text{admin\_hours} / \text{team\_size}) / \text{weeks\_per\_year}$$

Normalises the annual figure back to a weekly per-person view. More legible for individuals.

Step 6 — Reclaimable hours per year [OUTPUT 3]

$$\text{reclaim\_hours} = \text{admin\_hours} * (\text{adoption\_pct} / 100) * 0.60$$

Applies the adoption rate and the reclaim coefficient (0.60) to the admin hours pool.

Step 7 — Value of reclaimed capacity [OUTPUT 4]

$$\text{reclaim\_cost} = \text{reclaim\_hours} * \text{cost\_per\_hour}$$

Converts reclaimable hours into an equivalent monetary value.

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The Five Outputs

The calculator surfaces four primary metrics and two visual severity indicators.

| Output                               | Formula ref. | Plain English meaning                                                 | Colour                       |
|--------------------------------------|--------------|-----------------------------------------------------------------------|------------------------------|
| Annual cost of coordination overhead | Step 4       | Total salary cost of admin overhead per year                          | Magenta #C4006A              |
| Hours lost per year                  | Step 3       | Total hours consumed by work about work annually                      | Magenta #C4006A              |
| Hours per person per week            | Step 5       | Individual weekly view of coordination overhead                       | Magenta #C4006A (Change 002) |
| Reclaimable hours per year           | Step 6       | Hours recoverable with current AI adoption rate                       | Green #006B2B                |
| Value of reclaimed capacity          | Step 7       | Salary-equivalent value of recovered capacity                         | Green #006B2B                |
| Admin load severity bar              | admin_pct    | Below 25%: green. 25% and above: magenta. Amber removed. (Change 003) | Green / Magenta              |
| AI reclaim potential bar             | adoption_pct | Always green: represents opportunity, not risk                        | Always green #006B2B         |

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## Worked Example: Default Inputs

The following traces every calculation using the tool's default values.

| Input                  | Value                |
|------------------------|----------------------|
| team_size              | 25                   |
| avg_salary             | €80,000              |
| admin_hrs (user input) | 14 hrs/week          |
| admin_pct (converted)  | $(14/40)*100 = 35\%$ |
| adoption_pct           | 50%                  |
| weeks_per_year         | 48 (fixed)           |
| hours_per_week         | 40 (fixed)           |

|                      |                                                                                      |
|----------------------|--------------------------------------------------------------------------------------|
| cost_per_hour        | $€80,000 / (40 \times 48) = €41.67 / \text{hr}$                                      |
| total_hours          | $25 \times 40 \times 48 = 48,000 \text{ hrs} / \text{yr}$                            |
| admin_hours          | $48,000 \times 0.35 = 16,800 \text{ hrs} / \text{yr}$                                |
| admin_cost OUTPUT    | $16,800 \times €41.67 = €700,056 / \text{yr} \approx €700\text{K}$                   |
| hours_pp_week OUTPUT | $(16,800 / 25) / 48 = 14.0 \text{ hrs} / \text{person} / \text{week}$                |
| reclaim_hours OUTPUT | $16,800 \times 0.50 \times 0.60 = 5,040 \text{ hrs} / \text{yr} \approx 5.0\text{K}$ |
| reclaim_cost OUTPUT  | $5,040 \times €41.67 = €210,005 / \text{yr} \approx €210\text{K}$                    |

"A 25-person team at €80K average salary, spending 14 hours per week on coordination overhead, is losing €700K of productive capacity per year. With 50% AI adoption, €210K of that is recoverable: equivalent to roughly 2.5 FTEs."

## Stress Test: What If a Stakeholder Challenges the Numbers?

### "16.4 hours seems high."

It is a published composite from Microsoft 365 telemetry (n = millions of users) and APQC survey data (n = 982 knowledge workers). The model defaults to 14 hours per week (35%), which is conservative relative to the benchmark. Invite the stakeholder to adjust the slider to their own estimate.

### "60% reclaim rate is optimistic."

It is the weighted midpoint of three conservative research ranges, each from peer-reviewed or large-scale experimental studies. It applies only to admin hours, not total hours. At 35% admin load, the net reclaim is 21% of total capacity, not 60%.

### "This assumes AI adoption actually works."

The adoption\_pct slider lets the user set their own assumption. At 0% adoption, reclaim = €0. The model makes no claim that AI adoption happens automatically: only that when it does, these are the measurable gains.

### "Salary data may not match our organisation."

The avg\_salary input is fully adjustable. The model uses fully-loaded cost by convention, which includes base salary, employer contributions, and benefits. Teams should substitute their own figure.

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## Known Limitations

The model is intentionally simplified. The following limitations should be communicated proactively in client contexts.

- It does not differentiate by role type. A legal team and a software team may have very different admin profiles.
- It uses a single salary figure for the whole team. In practice, seniority varies and so does the cost of an hour.
- It does not account for the cost of AI tool implementation, training, or change management.
- It assumes linear scaling. In reality, reclaim gains may plateau above certain adoption thresholds.
- It is a planning and prioritisation tool, not an audited financial model.

*These limitations are features, not bugs: they create the conversation that leads to a consulting engagement.*

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## Sources

- Microsoft 365 telemetry: meeting and communication time data
- APQC: knowledge worker benchmark survey (n=982)
- Noy and Zhang (MIT): generative AI productivity study
- Harvard Business School: Navigating the Jagged Technological Frontier